UPI Remitter Bank Reliability Analysis

Is UPI technical reliability improving, and which banks should we watch first?

NPCI Top 50 remitter banks

Project report by Sonu, Nov 2026

24 months of data (Aug 2024 to Jul 2026)	**1,198** bank-month rows analysed	**49** banks with 12+ months, ranked and compared	**0.43% to 0.32%** technical decline rate, year 1 to year 2	**5 of 9** findings hold under our own stress tests

In plain: out of every 1,000 UPI payments from these banks, about 4 failed in year 1 because of a bank-side technical problem, and about 3 in year 2.

The story in 30 seconds

- **It is improving.** Technical declines fell from 0.43% to 0.32% of payments year on year, while volume grew 29%. The number of failed payments stayed roughly flat (down 3%) on much more traffic.
- **But the last five months rose** (0.20% in Feb 2026 to 0.40% in Jul 2026). Swings of 0.1 to 0.2 points are normal here, so this is a watch item, not yet a trend.
- **A few banks matter most.** 6 of 49 banks account for about **55%** of technical failures. State Bank of India alone is about 34%: it is the biggest bank, and its failure rate is also 1.5 times that of other large banks.
- **Small banks lag.** They handle 6% of payments but produce 18.5% of failures, a rate 3.6 times that of large banks.
- **The prize is large, but it is a ceiling.** If every weaker bank matched its size peers, up to 62% of technical failures could be avoided. A stricter reading gives 41% to 50%.
- **We tested ourselves.** 5 of 9 findings survive our own stress tests. 4 need a caveat, and the report says which.

1. Objective and business question

Decision question: a UPI app team, or a bank's operations team, asks:

→**Is UPI technical reliability improving, and which banks should we watch first?**

What this project delivers:

(1) the trend over 24 months,

(2) a short watchlist of banks,

(3) a fair comparison between banks of similar size,

(4) the size of the improvement prize,

(5) an honest view of how solid each finding is. Every finding says "associated with", never "caused by".

A 60-second UPI primer

Outcome of a payment	Share (these banks)	What it means in plain terms
Approved	91.3%	The payment went through.
Business decline (BD)	8.3%	Failed for a customer-side reason, for example wrong PIN or not enough balance. Not the bank's technical fault.
Technical decline (TD)	**0.37%**	Failed because of a technical problem on the bank's side. **This is the metric we study.**

In plain English: 0.37% sounds tiny, but UPI is huge. Across these 50 banks and 24 months, about 490 billion payments were attempted, and an estimated **1.82 billion** failed for technical reasons. That is why a few hundredths of a percentage point matter.

2. Tech stack and data

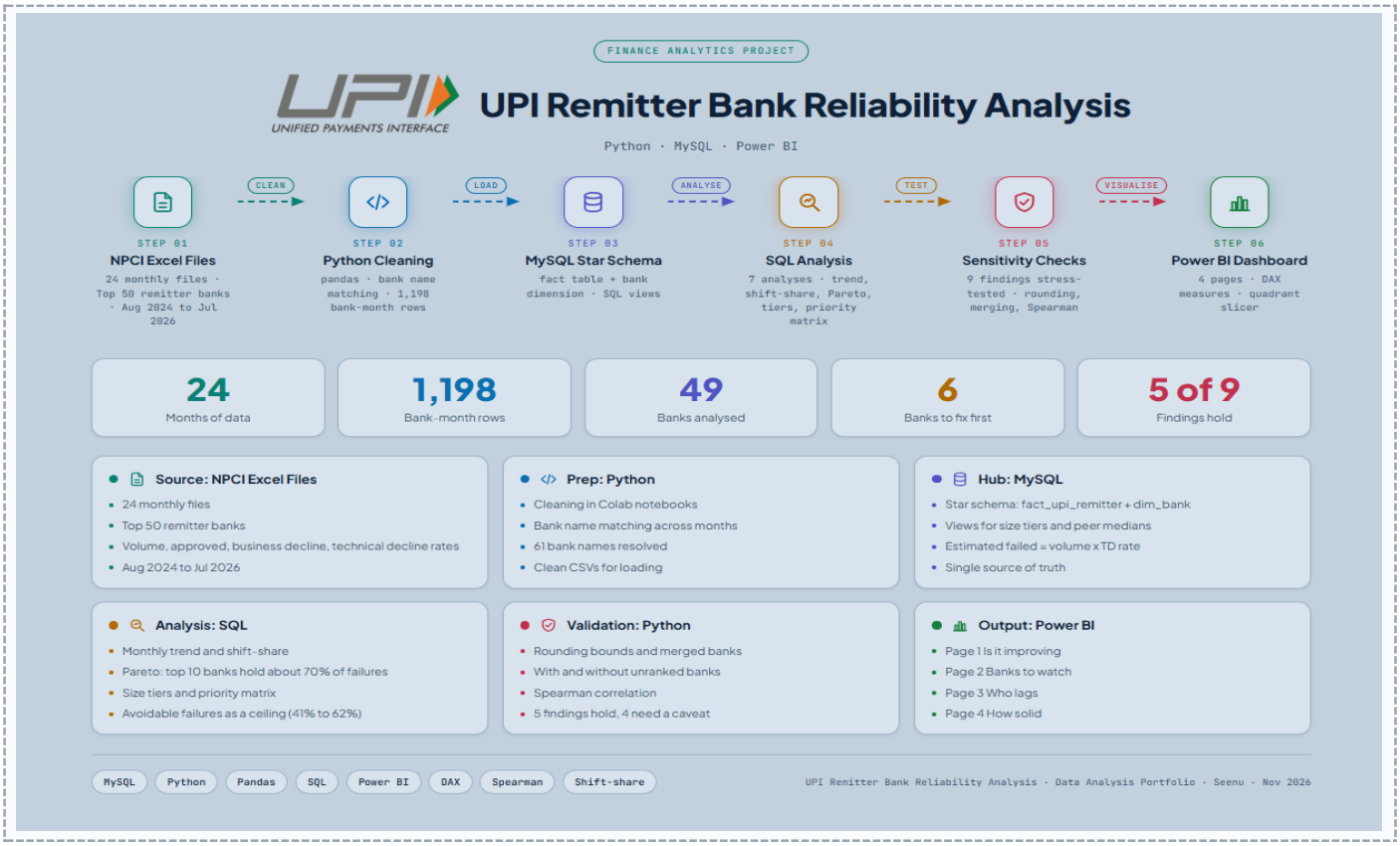
Stage	Tools	What it did
Collect	NPCI website (Ecosystem Statistics, Top 50 member performance, Remitter tab)	24 monthly Excel files, one per month, downloaded manually
Clean	Python (pandas, openpyxl)	Standardised 3 different file layouts, validated each file, matched bank names
Store	MySQL (Workbench)	Star schema: one fact table (bank x month) and a bank dimension, plus views
Analyse	SQL (CTEs, window functions, views), Python (scipy)	7 analyses, 9 stress tests, and the Spearman correlations
Visualise	Power BI (DAX, star model)	4-page dashboard reading the SQL views, every number reconciled to SQL
Version	Git and GitHub	Full code, SQL and dashboard file in one repository

What the data looks like

- **Source:** NPCI "UPI Remitter Members, Top 50", Aug 2024 to Jul 2026. Each month lists 49 or 50 banks with volume (millions of payments) and the approved, BD and TD rates.
- **Three different file layouts:** rates stored as fractions in some files, as text with "%" and commas in others, plus extra blank columns and different sheet and column names. All were converted to one clean table of **1,198 bank-month rows**.
- **Bank names were messy:** 117 name variants resolved to **61 banks** (85 spelling variants fixed, 6 names tied to a renamed or merged bank, 3 names kept separate because a merger is likely but not proven, 23 already clean). 6 entries that are not regular banks (4 credit-card entities, a wallet company, a payments firm) are kept in the data but left out of bank rankings.
- **Checks run before any analysis:** approved + BD + TD adds up to 100%, no duplicate bank-month rows, loader stops on any surprise, row counts match between raw staging and final tables, and a hand check against the source files.

> **In plain English:** if "State Bank Of India" and "State Bank of India" were counted as two banks, SBI's share of failures would be split in half and the ranking would be wrong. Cleaning first, analysis second.

3. Methodology and workflow



#	Business question	Method	What it produced
1	Is it improving?	Time series, month-on-month change in percentage points	Monthly trend, 24 points
2	Real improvement, or just traffic moving?	Shift-share (rate-mix) split on a balanced panel of 39 banks	Rate effect -0.132 pp, mix effect +0.015 pp
3	Who matters most?	Pareto and concentration ratio (CR5)	Top 10 banks = 71% of failures
4	Chronic problem or a bad month?	Persistence and volatility (coefficient of variation)	4 profile types
5	Is the comparison fair?	Size-tier benchmarking (Large, Mid, Small)	Peer median per tier per month
6	What do we fix first?	Priority matrix: impact vs severity	6 "Fix first" banks
7	How big is the prize?	Counterfactual (what-if) vs peer median	41% to 62% ceiling
8	Does heavy traffic cause failures?	Spearman correlation on month-on-month changes	No reliable link
+	Do results survive our own choices?	Sensitivity analysis (rounding, merged banks, with and without unranked, gentler targets)	9 checks, 5 hold

Choices we made, and say so openly

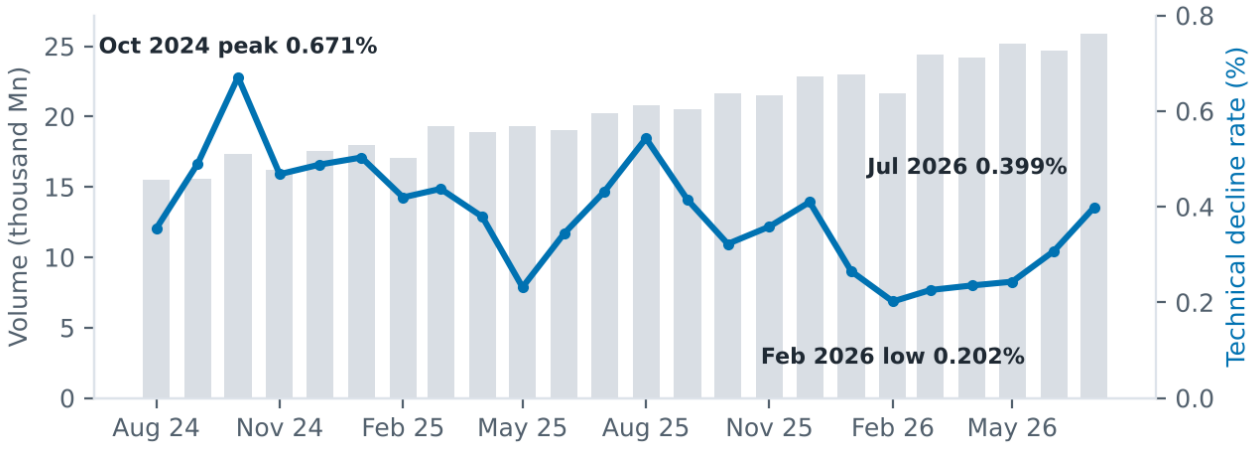
- **Eligible banks:** ranked banks with 12 or more months of data = 49. **Size tiers** by average monthly volume: Large 500 Mn or more (13 banks), Mid 100 to 500 Mn (11), Small under 100 Mn (25).
- **Peer median** = the median technical decline rate of banks in the same tier in the same month. A bank is "worse than peers" when it is above that line.
- **Year-on-year uses 12 months against 12 months** (Aug 2024 to Jul 2025 vs Aug 2025 to Jul 2026). A single-month comparison flipped sign depending on the months chosen, so it is kept only as a check.
- **Bank averages are total failed / total volume**, not an average of monthly rates, so a bank's busy months count more.

4. Key KPIs and what they mean

KPI	How it is calculated	What it means in plain English
Weighted TD rate	Total estimated failed / total volume	Share of payments that failed for technical reasons. Big banks count for more, as they should.
Estimated failed (Mn)	Volume x TD % / 100	NPCI publishes rates, not failed counts, so these counts are estimates.
Percentage points (pp)	Difference between two rates	0.43% to 0.32% is a fall of 0.11 pp, which is about 25% in relative terms.
Impact %	A bank's failures / all banks' failures	How much of the problem sits with this bank. If shared evenly, each of 49 banks would hold 2.04%.
Severity (x)	Bank rate / its size tier's rate	1.0 = same as similar-size banks. 2.0 = twice as bad. "Fix first" needs 1.25 or more.
Persistence %	Share of months worse than the tier median	Chronic (75% or more) vs one-off. Shows whether a problem is a habit.
Coefficient of variation	Spread of the monthly rate / its average	High means spiky and unpredictable, low means steady.
Technical share of declines	TD failures / (BD + TD failures)	Of a bank's failed payments, how many were the bank's own technical fault.
Avoidable failures	Failures above the tier median, added up	A ceiling for improvement if weaker banks matched their peers. Not a forecast.

5. Findings, each with its plain-English story

Finding 1: Technical reliability improved year on year, but has risen for five months



Grey bars: volume. Blue line: weighted technical decline rate.

Number	Value	Meaning
Rate, Year 1 (Aug 2024 to Jul 2025)	0.432%	Starting point
Rate, Year 2 (Aug 2025 to Jul 2026)	0.324%	Down 0.108 pp, about 25% lower
Volume, Year 2 vs Year 1	+29%	214,121 Mn to 276,276 Mn payments
Estimated failed, Year 1 to Year 2	925 to 895 Mn (-3%)	Failures flat on 29% more traffic
Balanced panel (39 banks) rate	0.414% to 0.297%	Of the -0.117 pp: banks improving -0.132, traffic mix +0.015
Peak, low, latest month	0.671% / 0.202% / 0.399%	Oct 2024, Feb 2026, Jul 2026

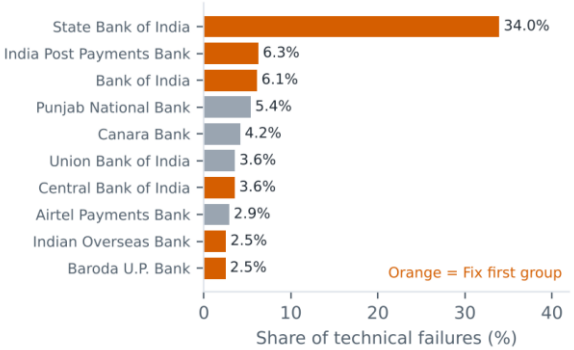
In plain English: the failure rate fell by a quarter while traffic grew by more than a quarter, so the system carried far more payments with about the same number of failures. The shift-share test shows the banks themselves got better: traffic drifting toward higher-rate banks (+0.015 pp) worked slightly against the improvement. Since Feb 2026 the rate has risen every month, but moves of 0.1 to 0.2 points are common, so five months is a watch item, not a trend. A "festive season effect" is not supported: Oct 2024 spiked (0.671%) but Oct 2025 was low (0.322%).

Finding 2: A handful of banks account for most technical failures

Concentration: on the 49 eligible banks, SBI holds **34.0%** of failures, the top 5 hold **55.9%** and the top 10 hold **71.0%**. (On all 55 ranked banks: 33.6%, 55.3%, 70.3%.) Month by month, the top 5 held between 51% and 77% (average 63%) with no steady drift.

Priority matrix: a bank goes in **Fix first** when it is both big (2.04% or more of failures) and weak (1.25 times its tier rate or more). Six banks qualify and together hold **54.9%** of failures, about 989 of 1,800 Mn.

In plain English: if failures were spread evenly each bank would own about 2%. Six banks own 55%. SBI is the largest source simply because it is the biggest bank, but it also fails 1.5 times as often as other large banks and is worse than its peers in 92% of months. Baroda U.P. Bank is tiny (0.2% of volume) but fails 3.6 times as often as banks its size.



Share of technical failures, top 10 of 49 banks

Fix first bank	Tier	Share of failures	Rate vs size peers	Months worse than peers	Technical share of declines
State Bank of India	Large	33.95%	1.54x	91.7%	6.4%
India Post Payments Bank	Large	6.28%	2.31x	91.7%	4.7%
Bank of India	Large	6.11%	2.37x	79.2%	9.3%
Central Bank of India	Mid	3.56%	2.34x	100.0%	7.3%
Indian Overseas Bank	Mid	2.54%	1.59x	66.7%	7.0%
Baroda U.P. Bank	Small	2.50%	3.60x	95.0%	40.6%

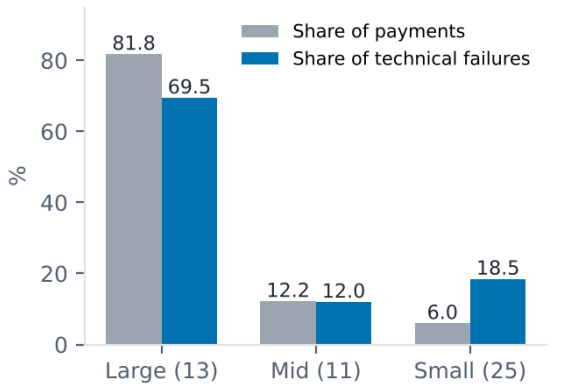
Finding 3: Small banks lag the most

Rates by tier: Large **0.315%**, Mid **0.365%**, Small **1.148%**. A small bank fails **3.6 times** as often as a large one, and small banks produce 18.5% of failures on 6.0% of volume.

Profile of the 49 banks: Healthy **19**, Chronic **13** (worse than size peers in 75% or more of months), Episodic **9** (usually fine but spiky), Mixed **8**.

Technical share of declines: Axis 0.5%, HDFC 0.7% and Kotak 1.0%, against 18% to 41% for the five rural banks with the highest share (Baroda U.P. 40.6%).

In plain English: comparing a small rural bank with HDFC would be unfair, so each bank is compared only with banks of similar size. Even so, small banks fail 3.6 times as often. About 1 in 5 technical failures comes from banks that handle about 1 in 17 payments. For HDFC and Axis almost none of the declines are their own fault; for the weakest rural banks, their own systems are a main reason payments fail.



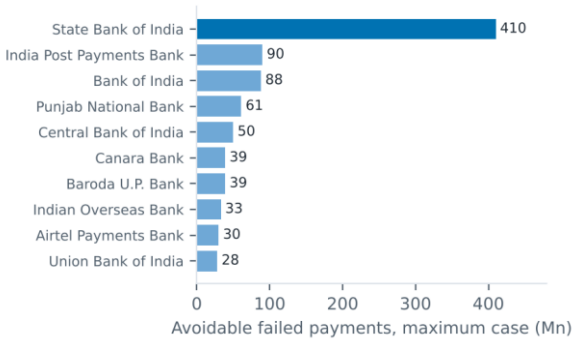
Share of payments vs share of technical failures, by size tier

Finding 4: Up to 62% of failures are avoidable, as a ceiling

What-if: if every bank-month worse than its size peers had matched the peer median, **1,118 of 1,800 Mn** estimated failures (**62.1%**) would not have happened.

Gentler targets: 56.0% if banks match 1.25 times the median, **50.4%** at 1.5 times, **41.0%** at 2 times. SBI alone is about **410 Mn**, over a third of the total.

In plain English: no bank can sit exactly on the median every month, so 62% is the most that could be saved, not a forecast. The safe way to say it is "roughly 40% to 60% of technical failures look avoidable", and the Fix first banks are where most of it sits.



Avoidable failed payments, top 10 banks (Mn, maximum case)

Finding 5: Heavy traffic does not reliably mean more failures

Test: *Spearman correlation* between month-on-month changes in volume and in the technical decline rate. National monthly: rho **0.39**, p = 0.065, n = 23 (weak, not significant at 5%). Bank level: rho **-0.01**, p = 0.84, n = 1,078 (no relationship).

In plain English: a busier month does not reliably mean a worse month, and across banks there is no link at all. So we do not claim that load causes failures. We used changes, not levels, because both series rise over time and levels would show a false link.

6. Stress tests: what held and what did not

Finding	Stress test	Result	Verdict
Year on year improvement (0.432% to 0.324%)	S1 rounding bounds	0.427 to 0.437 vs 0.319 to 0.329	**Holds**
Fix first banks run 1.25x or more their tier rate	S1 worst case	weakest case 1.50x	**Holds**
Same 6 banks in Fix first	S2 merge 3 pairs	no change	**Holds**
Top 10 hold about 70% of failures	S2 merge 3 pairs	70.3% to 70.7%	**Holds**
Trend unaffected by unranked entries	S3 with and without unranked	largest monthly gap 0.0052 pp	**Holds**
Avoidable failures (62% at median)	S4 gentler target	50% at 1.5x, 41% at 2x	**Ceiling only, quote a range**
Episodic label	S5 minimum rate	9 banks, 7 with a 0.10 floor	**Partly an artifact of tiny rates**
Volume vs TD rate, national monthly	Spearman on MoM changes	rho 0.39, p 0.065, n 23	**Weak, not significant at 5%**
Volume vs TD rate, bank level	Spearman on MoM changes	rho -0.01, p 0.84, n 1,078	**No relationship**

In plain English: we tried to break our own findings. The headline results (improvement, the six banks, the 70% concentration) survive rounding and merged banks. The avoidable-failure figure is only a ceiling. The "Episodic" label is partly an artifact: a bank with a near-zero rate looks spiky from tiny changes. And there is no reliable link between traffic and failures. Reporting the weak ones is the point.

7. The Power BI dashboard

Four pages, one message each. The data model is a star schema (one fact table of bank-month rows, bank and date dimensions) plus two SQL views for bank and tier results. Every KPI was reconciled to a verified SQL value before use.

Page	Headline message	Key numbers
1 Is it improving	Technical declines fell from 0.43% to 0.32% year on year, while UPI volume grew 29%.	0.40% latest month, -0.11 pp, +29%, -3%
2 Banks to watch	6 of 49 banks account for 55% of technical failures. State Bank of India alone is 34%.	6, 54.94%, 33.95%, 71.03%
3 Who lags	Small banks fail 3.6 times as often as large banks. Matching size peers could avoid up to 62% of failures.	3.65x, 18.51%, 62.12%, 1,118 Mn
4 How solid	5 of 9 findings hold under our own stress tests; 4 need a caveat.	9, 5, 4, 1,198 rows

8. Limits

Limits

- **Top 50 banks only.** Not the national total.
- **Rates are rounded and failed volumes are estimates** (volume x TD rate).
- **Avoidable failures are a ceiling,** not a forecast.
- **Banks with under 24 months** (Andhra Pragathi 14, Telangana 19, Baroda U.P. 20) may be a biased sample.
- **"Episodic" is partly an artifact** of near-zero rates.
- **No reliable volume-to-rate link** at bank level.
- **NPCI's 1% technical decline benchmark is unconfirmed,** so it is not used.

Appendix A:

Question	Short answer
Why a weighted rate and not a simple average?	A simple average would let a tiny bank count as much as SBI. Weighting by volume gives the real share of payments that failed.
Why 12 months vs 12 months?	The Aug 2024 vs Jul 2026 single-month comparison flipped sign depending on the months chosen. 12 vs 12 is stable.
What is shift-share, and what did it show?	It splits a change into "banks improved" and "traffic moved". Banks improving gave -0.132 pp, traffic mix +0.015 pp, on 39 banks present all 24 months.
Is "62% avoidable" real?	It is a ceiling: no bank can sit on the peer median every month. A fair statement is a range, 41% to 62%.
Why correlate changes and not levels?	Volume and the rate both trend over time, so levels would show a false link. Changes did not (rho 0.39, not significant nationally; none at bank level).
What would make you less confident?	Top 50 only, rounded rates, estimated failures, a few banks with short history, and an unconfirmed NPCI benchmark. The 4 caveats are listed on page 4 of the dashboard.
What did you decide, and what did data decide?	I set the tiers, the 1.25x severity bar, the 75% chronic bar and the 12-month rule. The data then picked the six banks, and the sensitivity checks kept the ranking unchanged.

Appendix B: Glossary

Term	Meaning
UPI / NPCI	Unified Payments Interface, India's instant payment system, run by the National Payments Corporation of India.
Remitter bank	The bank that sends the money, the customer's own bank at the time of paying.
BD / TD	Business decline (customer-side reason) and technical decline (bank-side technical problem).

Term	Meaning
Mn / pp	Million payments / percentage points (difference between two percentages).
Tier	Size group by average monthly volume: Large, Mid, Small.
Shift-share	Splits a change in a total rate into improvement inside banks and a shift of traffic between banks.
Pareto / CR5	A few items hold most of the total; CR5 is the share held by the top 5.
Spearman correlation	A rank-based link between two series; near 0 means no relationship.
Star schema	A fact table of numbers surrounded by small tables that describe them (bank, date).